

1	Define an Intrusion Detection System (IDS).
2	List two main types of IDS.
3	State any two goals of IDS.
4	What is the primary function of an IDS?
5	Differentiate between HIDS and NIDS.
6	Mention any two internal threats to data.
7	Explain the term 'structured threat' with an example.
8	What is a false positive in IDS?
9	State the function of E-box in IDS architecture.
10	Name two audit record fields.
1	What is meant by audit logs in IDS?
2	List any two requirements of an effective IDS.
3	Give two examples of masqueraders.
4	Mention any two advantages of misuse detection.
5	Explain 'profile-based anomaly detection' with an example.
6	What is the effect of base-rate fallacy in IDS?
7	How does IDS assist in forensic analysis?
8	Define a 'signature' in misuse detection.

9	What is an R-box in IDS architecture?
10	Explain the impact of threshold based detection in IDS
1	What is the primary function of an Intrusion Prevention System (IPS)?
2	Differentiate between IDS and IPS.
3	List the four components of a typical IPS architecture.
4	State two key benefits of using IPS in enterprise security.
5	What is Host-Based IPS (HIPS)?
6	Mention two malicious behaviors that HIPS can detect.
7	Define Network-Based IPS (NIPS).
8	What is flow data protection in NIPS?
9	Mention two components of NIDS.
10	What is the role of the detection engine in NIDS?
1	Where is NIDS usually deployed for monitoring DMZ traffic?
2	What is the advantage of placing NIDS outside the firewall?
3	Differentiate between Atomic and Composite pattern descriptors.
4	What is Stateful Protocol Analysis?
5	What is the first step in intrusion analysis?
6	What is the role of feature extraction in intrusion analysis?

7	What is meant by rate limiting in IDS response?
8	What is vulnerability analysis?
9	List the two classifications of vulnerability analysis tools.
10	What is credential vulnerability analysis?
1	Explain the use of the content keyword.
2	What is the purpose of the flags keyword in Snort?
3	Differentiate between -> and <> in Snort rules.
4	What is the significance of classtype in Snort rules?
5	What is the use of sid in Snort rules?
6	Mention the importance of rev in Snort rules.
7	What are the advantages of using variables in Snort configuration?
8	Explain the function of the dsize keyword.
9	What is the role of the itype and icode keywords?
10	Explain frag2 preprocessor.
1	Explain stream4 preprocessor.
2	What is portscan detection in Snort?
3	What is the difference between alert and log actions in Snort?
4	Define Oinkmaster.

5	What is the use of the include keyword in snort.conf?
6	What are the categories of default Snort rule files?
7	How is database output used in Snort?
8	What is the role of Stunnel in Snort logging?
9	Differentiate between alert_fast and alert_full output plugins.
10	What is the purpose of Berkeley Packet Filters (BPF) in Snort?
1	Differentiate between raw sockets and pcap API in sniffing.
2	What is packet spoofing?
3	Mention two tools for packet sniffing.
4	Explain TCP reset attack.
5	State the significance of the Mitnick attack
6	What is DNS cache poisoning?
7	What is the Kaminsky attack?
8	What is DNSSEC?
9	How does a denial-of-service attack affect DNS servers?
10	Define DNS rebinding attack.
1	What is TCP session hijacking?
2	What is the TCP three-way handshake?

3	What is a SYN flooding attack?
4	How does sniffing help in spoofing?
5	What is an ICMP redirect attack?
6	How does ARP poisoning enable man-in-the-middle attacks?
7	Which is called as Swiss Army Knife
8	What are the various purposes of netcat command
9	How loopback address is used for trouble shooting
10	How to perform sniffing using tcpdump
1	What is the main role of a VPN server in a VPN setup?
2	Compare IPsec Tunneling and TLS/SSL Tunneling.
3	List the five IPv4 Netfilter hooks.
4	What is the difference between ingress and egress filtering?
5	Define a firewall.
6	Give an example of an iptables rule to drop incoming ICMP echo requests
7	Define Virtual Private Network (VPN).
8	How can the VPN tunnel be tested using ping and tcpdump?
9	Explain reverse SSH tunneling with an example.
10	How can other hosts use a dynamic SSH tunnel?

1	What is an autonomous system (AS), and why is it important in BGP routing?
2	Explain the concept of “longest prefix match” in BGP routing and how it relates to IP prefix hijacking.
3	What is a reverse shell, and why is it commonly used in hacking?
4	Explain the role of file descriptors in the functioning of a reverse shell.
5	How does the dup2() system call help in redirecting I/O for a reverse shell?
6	In a real attack, why is the reverse shell command often passed to another shell using -c?
7	What is the role of the SOCKS protocol in dynamic port forwarding?
8	How does the VPN server release packets into the private network?
9	What are TUN and TAP interfaces in VPNs?
10	List two main functions of a firewall.

Part B

Unit 1		
1	i)	Describe the evolution of intrusion detection and explain how auditing has shaped the development of IDS.
	ii)	Explain how auditing practices have influenced the development of intrusion detection systems throughout history.
2	i)	Compare internal threats and external threats to data, giving appropriate attack examples.

	ii) Illustrate with examples how internal threats differ from external threats to data security.
3	i) Trace the historical development of intrusion detection and explain the importance of auditing in this process. ii) Explain How does the concept of auditing support the development of intrusion detection systems?
4	Classify IDS according to their deployment location and briefly explain each type.
5	Explain the taxonomy of intrusion detection systems (IDS) and categorize them based on detection methods.
6	Differentiate IDS taxonomy in terms of anomaly-based detection and signature-based detection with examples.
7	Describe in detail the architecture of an Intrusion Detection System (IDS) and explain the function of each component.
8	Discuss the major goals of an IDS and explain how these goals enhance network and system security.
9	Explain the significance of auditing in intrusion detection systems and how audit data is utilized to detect attacks.
10	i) List and briefly describe the key components of IDS architecture. (7 Marks) ii) Explain how these components work together to detect intrusions. (6 Marks)
1	Compare rule-based IDS techniques with anomaly-based techniques in terms of detection capability and efficiency.
2	Compare the advantages and disadvantages of misuse-based detection when compared to anomaly-based detection.
3	Examine the effectiveness of rule-based IDS in detecting known and unknown attacks.
4	Examine how anomaly-based detection helps in identifying zero-day attacks, and

	discuss its limitations.
5	i) Compare misuse-based detection with rule-based detection techniques, highlighting scenarios where each is more effective.
6	i) Examine the challenges of combining anomaly-based and misuse-based detection in a hybrid IDS.
7	Compare host-based and network-based information sources in IDS and analyze their role in intrusion detection accuracy.
8	Examine the challenges in using network-based information sources for large-scale IDS deployment.
9	Demonstrate how combining host-based and network-based information sources enhances the effectiveness of IDS.
10	i) Examine the strengths of host-based information sources in intrusion detection. (7 Marks) ii) Discuss the limitations of host-based sources when compared to network-based sources. (6 Marks)
Unit 2	
1	Explain the working principle of a network-based intrusion detection system (NIDS).
2	Describe the main functions of a host-based intrusion detection system (HIDS).
3	Compare host-based and network-based IDS in terms of their detection scope and deployment.
4	i) List and explain the advantages of network-based IDS. (7 Marks) ii) Describe the limitations of network-based IDS. (6 Marks)
5	i) Explain how host-based IDS monitors system logs and files. (7 Marks) ii) Discuss the disadvantages of using only host-based IDS. (6 Marks)

6	i) Describe the situations where host-based IDS is preferred over network-based IDS. (7 Marks) ii) Give examples of cases where network-based IDS is more effective. (6 Marks)
7	Explain the working of a protocol-based intrusion detection system (PIDS) with an example.
8	Describe the features and benefits of a hybrid intrusion detection system (Hybrid IDS).
9	What is an Intrusion Prevention System (IPS)? Explain how it differs from IDS.(7 marks) Compare IDS vs IPS vs Firewall(6 marks)
10	i) Explain how protocol-based IDS monitors network protocols for suspicious activity. (7 Marks) ii) Discuss the limitations of protocol-based IDS. (6 Marks)
1	Examine the importance of intrusion analysis schemes in improving IDS effectiveness.
2	Compare different approaches to “thinking about intrusion” and discuss how they help in proactive security.
3	Demonstrate how a model for intrusion analysis supports decision-making in security operations.
4	Critically examine the challenges of applying structured analysis schemes to real-world intrusions.
5	i) Explain the key steps involved in a typical intrusion analysis model. (7 Marks) ii) Analyze how this model helps in identifying the source and nature of attacks. (6 Marks)

6	i) Discuss the role of “thinking about intrusion” in anticipating attacker strategies. (7 Marks) ii) Examine how security teams can apply this concept in real-time analysis. (6 Marks)
7	Examine the requirements of IDS responses such as active and passive and explain why timely responses are critical in intrusion detection.
8	Compare the different types of IDS responses and evaluate their effectiveness in mitigating attacks.
9	Examine how mapping IDS responses to security policy improves the overall defense mechanism of an organization.
10	Compare credential-based and non-credential-based vulnerability analysis, highlighting their advantages and limitations.
Unit 3	
1	(a) Analyze the procedure to install Snort using RPM package. (6) (b) Explain how to install Snort from source code. (7)
2	Demonstrate the working of Single Sensor with Database and Web Interface installation.
3	(a) Show with commands how Snort is run on multiple network interfaces. (6) (b) Justify why different log directories and configuration files are required. (7)
4	Examine and analyze the working of Multiple Sensors with Centralized Database installation. Why might SCP be preferred over real-time logging?
5	(a) Demonstrate with commands how Snort can be started in foreground mode and daemon mode. (6) (b) Discuss the role of the snortd startup script and runlevel directories. (7)

6	Illustrate with examples the use of Snort command line options: -c, -i, -l, and -D.
7	Analyze the configuration process of Snort after installation — examine how its directories, rule files, and configuration files interact to ensure accurate intrusion detection.
8	Analyze why testing Snort with ICMP echo packets is important. Explain how Snort interprets such traffic and generates alerts, highlighting what this reveals about its detection accuracy.
9	Analyze how compiling Snort with optional modules like MySQL, SNMP, and OpenSSL enhances its functionality. Interpret the purpose of each support feature and justify the use of specific ./configure options in the build process.
10	Explain the two major modes of Snort operation with examples.
1	Illustrate the different Snort alert modes: Fast, Full, and Syslog.
2	(a) Write short notes on UNIX socket alert mode and SNMP trap mode. (6) (b) Explain how alerts can be sent to Windows machines. (7)
3	Compare Snort foreground execution with daemon/background execution using suitable commands.
4	(a) Define the components of a Snort Rule Header with an example. (6) (b) Explain the purpose of Rule Options in a Snort rule. (7)
5	Explain with an example the structure of a basic Snort rule. Why is the rule alert ip any any -> any any considered a “bad” rule?
6	Analyze the importance of CIDR notation, addresses, ports, and direction fields in Snort rule headers with examples.
7	(a) Explain the working of Fast alert mode and Full alert mode in Snort with commands. (6)

	(b) Discuss the purpose of Syslog alert mode in Snort. (7)
8	(a) Describe the use of SNMP trap mode and UNIX socket mode in Snort alerts. (6) (b) Explain how alerts can be sent as Windows pop-ups. (7)
9	Explain any five important Snort rule option keywords such as content, ttl, msg, flags, and classtype with examples.
10	Differentiate between Snort installed via RPM package and Snort installed from source code.
Unit 4	
1	Analyze the life cycle of a packet across different layers of the protocol stack. How can attackers exploit each layer during transmission?
2	Explain the structure of an Ethernet frame. Analyze how ARP works and demonstrate how ARP cache poisoning enables man-in-the-middle (MITM) attacks.
3	With suitable examples, analyze vulnerabilities at the MAC layer. Discuss how MAC spoofing and ARP poisoning differ in execution and impact.
4	(a) Discuss the role of promiscuous mode in packet sniffing. (b) Evaluate how sniffing combined with spoofing can lead to session hijacking.
5	Compare the strengths and weaknesses of IP addressing and routing mechanisms. Analyze how IP fragmentation and TTL fields are misused in attacks.
6	Evaluate the role of IP header fields (TTL, fragmentation, and routing information) in maintaining packet delivery. How are these fields exploited to launch evasion and DoS attacks?
7	Examine the journey of a packet from application to physical layer. How can attackers misuse vulnerabilities at different stages of this process?

8	Describe the components of an Ethernet frame. Critically analyze how forged ARP replies compromise communication integrity in a LAN.
9	Assess the security challenges of the MAC layer. How do attacks like MAC address spoofing and ARP cache manipulation threaten confidentiality and availability?
10	(a) Explain how enabling promiscuous mode in a NIC aids packet capture. (b) Analyze how attackers combine sniffing with spoofed packets to disrupt or hijack sessions.
1	Describe the TCP three-way handshake. Explain how attackers exploit this mechanism to launch a SYN flooding attack.
2	Analyse the operational impact of a TCP reset (RST) attack: explain how a forged RST (using the RST flag with appropriate seq/ack values) forces endpoints to tear down an established TCP state and list the main effects on availability and session integrity for long-lived connections (e.g., SSH, streaming).
3	Describe the Mitnick attack as a case study: outline the steps (IP spoofing + TCP sequence-number prediction + injected packets) showing how predictable ISN allowed session takeover. Finally, name two effective defenses (randomize ISNs / use TCP-AO or SSH and deploy ingress/egress filtering).
4	With an example, explain the process of packet sniffing. How do tools like tcpdump or Scapy enable attackers to capture sensitive information?
5	Illustrate how packet spoofing is carried out using raw sockets. Explain with an example of an ICMP spoofing attack.
6	Analyse a TCP RST (Reset) attack and its detection
7	Analyse TCP session hijacking and forensic identification
8	Analyse the mechanics and mitigations of SYN flooding
9	Analyse and design a controlled lab to simulate a SYN flood

10	An organization is under a combined attack of SYN flooding and ARP poisoning, disrupting network connectivity. Design a Snort rule and preprocessor approach to detect both attacks simultaneously while minimizing false positives and ensuring comprehensive coverage.
Unit 5	
1	Design a Linux firewall using Netfilter that enforces custom security policies for a small office network. Justify your design choices and demonstrate how it handles both allowed and blocked traffic scenarios.
2	Analyze a given network traffic scenario and configure iptables rules to optimize security while maintaining network functionality. Evaluate the effectiveness of your rules against potential attacks.
3	Design and implement a stateful firewall using connection tracking in Linux. Compare its performance and security advantages against a stateless firewall, citing specific traffic examples.
4	Construct a comprehensive iptables rule set for a multi-service server. Justify your selection of rules and assess how well it mitigates different types of network attacks.
5	Develop a testable Linux firewall using Netfilter for a hypothetical network. Evaluate its ability to block unauthorized access while allowing essential services, and suggest improvements.
6	Critically analyze the core components and functions of firewalls. Compare their effectiveness in different network scenarios and propose an optimized firewall architecture for a corporate environment.
7	Evaluate various types of firewalls (packet filtering, stateful, application, and next-gen) in terms of performance, security, and scalability. Recommend the most suitable firewall type for a high-traffic web service and justify your choice.
8	Given a network with multiple departments and external connections, design a firewall deployment plan. Analyze which firewall type(s) best suit each segment and justify your selection based on security and performance trade-offs.

9	Compare TUN and TAP interfaces in the context of a VPN deployment. Analyze their advantages and limitations for different OSI layers, and recommend the most suitable interface for a corporate remote access scenario.
10	Design a secure VPN setup for remote employees. Analyze how encryption, tunneling, and authentication mechanisms protect data, and evaluate potential vulnerabilities.
1	Analyze how file descriptors, I/O redirection, and TCP connections enable reverse shell creation. Design a scenario showing how these mechanisms interact, and evaluate potential defenses against such attacks.
2	Develop a C program using dup2() to redirect standard input/output to a network socket. Critically analyze how this technique facilitates reverse shell creation and propose mitigation strategies.
3	Design a stepwise demonstration showing how I/O redirection enables a reverse shell in bash. Evaluate how each stage contributes to the attack and propose measures to prevent exploitation.
4	Construct a controlled scenario using /dev/tcp in bash to initiate a reverse shell. Analyze the role of each command and assess possible defense mechanisms for both attacker and victim machines.
5	Design a reverse shell command using /dev/tcp for a system without netcat. Critically evaluate each component of the command and propose preventive measures to block such attacks.
6	Compare Heartbleed and SQL injection in terms of data exposure and impact on encrypted communications. Evaluate which vulnerability poses a greater threat to secure systems and justify your reasoning.
7	Analyze the mechanism by which Heartbleed allows server memory leakage. Assess the potential consequences of private key compromise and design a mitigation strategy to prevent exploitation.
8	Construct a detailed attack flow diagram for Heartbleed, showing each stage from the malicious heartbeat request to data extraction. Evaluate potential defenses

	and justify their effectiveness.
9	Design a post-Heartbleed recovery plan for a web server. Analyze each step, from patching OpenSSL to credential protection, and evaluate its effectiveness in restoring system security.
10	Develop a comprehensive incident response strategy for suspected Heartbleed exposure. Critically assess methods to detect leaked data and design long-term measures to prevent recurrence.

Part C

1	An organization wants to protect a highly distributed cloud environment. If you were the security analyst, which type of IDS (host-based, network-based, protocol-based, or hybrid) would you recommend and why?
2	A company reports frequent false positives from its IDS. As a system designer, suggest architectural improvements to reduce false positives and justify your approach.
3	A company experiences repeated port scanning on its external network. If you were the security engineer, how would you configure and evaluate a network-based IDS to detect and mitigate this threat?
4	An ISP needs to monitor massive volumes of traffic across multiple regions. Propose how a network-based IDS can be designed to scale efficiently while maintaining detection accuracy.
5	Explain how DoS attacks against DNS servers are performed. How do attacks on root servers differ from attacks on authoritative domain servers?
6	With examples, describe how reverse DNS lookup forgery can be misused by attackers to disguise malicious hosts as legitimate ones.

7	A system administrator observes abnormal latency and packet drops in a corporate network. On analysis, several packets show overlapping IP fragments. Analyze this situation and explain which IP fragmentation attack is likely happening. Suggest suitable countermeasures.
8	In a university LAN, students complain of frequent disconnections while accessing remote SSH servers. Logs show repeated TCP RST packets injected into sessions. Analyze the type of TCP attack occurring and discuss its impact on long-lived connections.
9	A service provider notices that users cannot reach a website because its Internet traffic is being sent through an unexpected foreign network. Question: Analyze what type of BGP-related problem this could be. Explain how it affects data flow and suggest one method that Internet providers can use to prevent this kind of attack.
10	A company sets up a VPN for remote workers, but some users report that even after connecting, their web traffic is still going directly through their home Internet instead of the company network. Question: Analyze why this situation happens. Explain one risk caused by this VPN setup and describe one simple change in the VPN configuration that would fix the issue.
1	A financial organization is concerned about zero-day attacks. If you were designing their IDS, which technique (rule-based, anomaly-based, misuse-based, or hybrid) would you apply, and why?
2	A university wants to monitor student network traffic without breaching privacy. Which IDS technique would you select, and how would you justify your choice?
3	An organization suspects insider misuse, such as unauthorized file access. As a security analyst, evaluate how a host-based IDS could be applied to detect and prevent such activities.
4	A critical application server is under threat of rootkits. Design a host-based IDS deployment plan and justify how it will protect the server.

5	Explain the working of a local DNS cache poisoning attack with a suitable example. How does it mislead users into connecting to malicious sites?
6	the remote DNS cache poisoning attack. How does the Kaminsky method improve the success rate of such attacks?
7	A company's internal router is misconfigured, allowing attackers to inject malicious ARP replies into employees' devices. Soon, sensitive login credentials are intercepted. Analyze this situation and explain how ARP cache poisoning enabled a man-in-the-middle (MITM) attack.
8	A popular news website suddenly becomes unreachable. Network analysis shows an overwhelming number of half-open TCP connections from spoofed IP addresses. Identify the attack, analyze how the TCP three-way handshake was exploited, and suggest mitigation strategies.
9	A firewall blocks all outgoing traffic except HTTP and HTTPS. Analyze how a VPN tunnel or SSH port-forwarding can bypass these restrictions, and discuss potential detection mechanisms.
10	A network admin allows all incoming traffic on port 22 for remote login. After a few days, there are many failed SSH login attempts from unknown IPs. Question: Analyze what mistake was made in the firewall configuration. Explain why it is risky and suggest how the rule can be modified to allow only secure access for administrators.